

Assignment 3: TLS Handshaking

Summary

Cryptographic ciphers are used to secure most communications in networking. Consequently, we need methods to relay keys to set up communication between a client and a server. This is the purpose of a TLS handshake. TLS allows us to add security above that of the Transport Layer in the TCP/OSI models where otherwise there would be unsecured communications.

Scenario

The Galactic Empire has been trying to cut us out of HoloNet communications for too long. Fortunately for us, a Bothan spy was able to get us the cipher suite that they use for their communications. Your job is straightforward. Using the document that our spy brought back, write a client that allows us access to the key exchange and then get an AES key that will be recognized on the HoloNet.

Objectives

You will be given a document that details the cipher suite used along with most of the code library used to connect to the server. You need to design the client that will.

- 1) Write to the key exchange to get an RSA key
- 2) Use the RSA key to then get an AES key
- 3) Use the AES key to access the HoloNet to get your unique flag.

Deliverables

The following are expected upon completion of this assignment.

- 1) You should be able to connect to the server and present your unique flag that's generated based on your name. A server-side log system will be retained to authenticate each student's interaction with the server.

Tips

- 1) Read the code provided closely. You've been provided with most of the code running on the server. Use that as you will.
- 2) The seeding code was designed to be hard to break, brute forcing the seed is not recommended.

Questions can be directed to chumme2@lsu.edu or Takonas on the LSU CSC discord